

# Phase 3 — Call Detection & Logging

## Design Document — Phoenix Computers Service Call App

**Version:** 1.0  
**Status:** Ready for Implementation  
**Depends on:** Phase 1 (Auth), Phase 2 (Task CRUD)  
**Estimated files:** 18 new + 6 modified

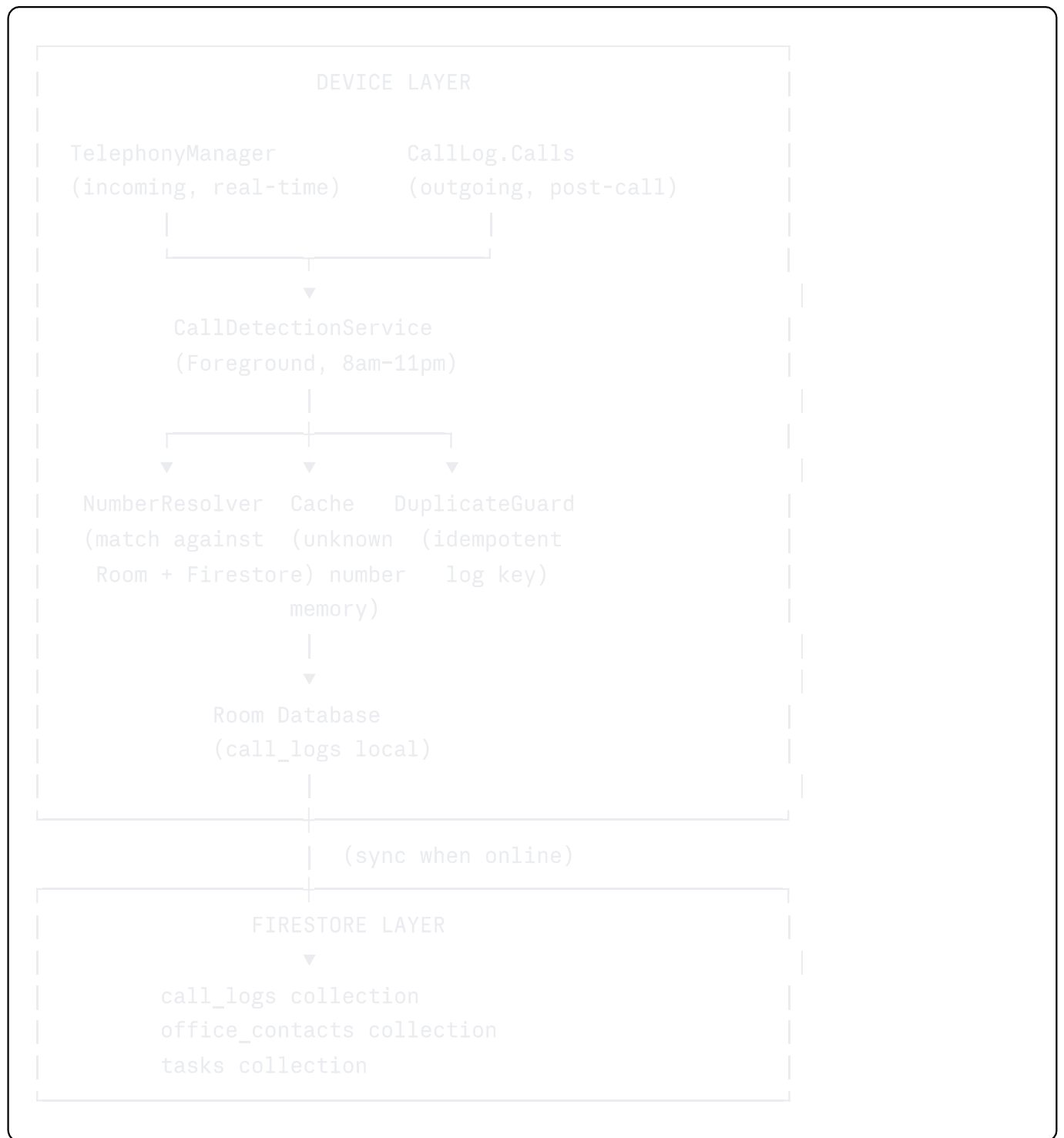
### 1. Objective

Automatically detect incoming and outgoing calls on each agent's device, match the caller against the shared office contacts list, and prompt the agent to log a service task — with minimal manual interaction, full offline resilience, and zero duplicate entries.

### 2. Scope

| In Scope                              | Out of Scope                        |
|---------------------------------------|-------------------------------------|
| Incoming call detection (real-time)   | SMS detection                       |
| Outgoing call detection (post-call)   | WhatsApp / VoIP calls               |
| Background service 8am-11pm           | Call recording                      |
| Office contact matching               | Group calls                         |
| Call bottom sheet (during/after call) | Phase 9 settings override           |
| My Calls tab (per-agent log)          | Voice notes (Phase 8)               |
| Missed call reminders (5 min)         | Service type intelligence (Phase 4) |
| Contact sync (phone → Firestore)      | Default dialer replacement          |
| Offline resilience via Room DB        |                                     |
| Auto-save when next call arrives      |                                     |

### 3. Architecture Overview



### 4. Call Detection — Dual Layer

#### 4.1 Layer 1 — TelephonyManager (Incoming, Real-Time)

**Class:** `CallStateReceiver.java`

**Trigger:** `PhoneStateListener` registered inside `CallDetectionService`

**Why not a manifest broadcast receiver:** Android 9+ restricts `PHONE_STATE` broadcast to foreground services. Registering inside the running service is the reliable modern approach.

## State machine:

```
IDLE → RINGING      : incoming call arrived → start timer, store number
RINGING → OFFHOOK   : call answered → mark as answered
RINGING → IDLE      : call missed → schedule MissedCallReminderWorker
OFFHOOK → IDLE      : call ended (answered then hung up) → show bottom sheet
```

**Reliability note:** On OEM ROMs (Samsung One UI, Xiaomi MIUI, OnePlus OxygenOS) the `PhoneStateListener` can be killed. This is caught by Layer 2.

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### 4.2 Layer 2 — CallLog Poller (Incoming Fallback + Outgoing Detection)

**Class:** `CallLogPoller.java`

**Trigger:** Heartbeat every 60 seconds inside `CallDetectionService`

**Reads:** `CallLog.Calls` — last 5 entries since last poll timestamp

**Stores:** `lastPollTimestamp` in `SharedPreferences`

#### What it catches:

- Outgoing calls (primary method for outgoing)
- Incoming calls missed by Layer 1 on restricted OEMs
- Missed calls not caught by Layer 1

**Cost note:** Reads only 5 local device records every 60 seconds. Zero Firestore reads. Negligible battery impact.

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### 4.3 Duplicate Prevention

**Strategy:** Composite key = `normalize(number) + floor(timestamp / 10000)`

This means two detections of the same call within 10 seconds produce the same key. The key is used as the Room DB primary key AND the Firestore document ID.

```
java
String dedupKey = normalizedNumber + "_" + (timestampMillis / 10000);
```

Write is idempotent — second write to same key is a no-op in both Room and Firestore.

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## 5. Service Lifecycle

### 5.1 CallDetectionService

**Type:** Foreground Service

**Notification:** "Phoenix Computers — Call detection active" (tapping opens MainActivity)

**Channel:** Existing (`CHANNEL_REMINDERS`) (reused, no new channel needed — cost saving)

**Start conditions (any of these):**

- Device boot → `BootReceiver` (already registered in Phase 1) checks if current time is 8am-11pm
- App launch → `MainActivity.onCreate()` checks time and starts if needed
- AlarmManager at 8:00am → `ServiceWindowReceiver`

**Stop conditions:**

- AlarmManager at 11:00pm → `ServiceWindowReceiver`
- Service stops itself if called outside 8am-11pm window

**What runs inside the service:**

- `PhoneStateListener` registration (Layer 1)
- `CallLogPoller` on 60-second `Handler` loop (Layer 2)
- `UnknownNumberCache` in-memory map (cleared at midnight)
- Reference to `RoomCallLogDao` for local writes

### 5.2 ServiceWindowReceiver

**Class:** `ServiceWindowReceiver.java`

**Registered:** In `AndroidManifest.xml` (exported, no permission needed for AlarmManager)

**Alarms set by:** `CallScheduler.scheduleWindow(context)` — called on boot and app launch

**Alarm type:** `AlarmManager.RTC_WAKEUP` with `setExact()`

```
Boot / App Launch
├─ CallScheduler.scheduleWindow()
│   └─ Cancel existing alarms
│   └─ Set exact alarm: next 8:00am → ACTION_START_DETECTION
│   └─ Set exact alarm: next 11:00pm → ACTION_STOP_DETECTION
```

**Battery note:** Two exact alarms per day. Android's exact alarm quota is not an issue at this frequency.

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## 6. Number Resolution

### 6.1 NumberResolver

**Class:** `NumberResolver.java`

**Input:** Raw phone number string

**Output:** `ResolvedContact` (name, category, contactId or null)

**Resolution order (cheapest first):**

1. `UnknownNumberCache` (in-memory) → instant, zero I/O
2. Room DB `office_contacts` mirror → local disk, no network
3. Firestore `office_contacts` (only if Room miss) → network call

**ResolvedContact categories:**

| Category                    | Meaning                         | Action                          |
|-----------------------------|---------------------------------|---------------------------------|
| <code>OFFICE_KNOWN</code>   | In <code>office_contacts</code> | Show sheet pre-filled           |
| <code>OFFICE_PENDING</code> | Marked office in cache today    | Show sheet pre-filled (no name) |
| <code>PERSONAL</code>       | Marked personal in cache today  | Log only, no sheet              |
| <code>UNKNOWN</code>        | Never seen before               | Show sheet, prompt category     |

### 6.2 UnknownNumberCache

**Class:** `UnknownNumberCache.java`

**Storage:** `HashMap<String, String>` in-memory (number → "office" or "personal")

**Cleared:** At midnight via a scheduled `Handler` post

**Persisted:** No — intentionally ephemeral. If service is killed and restarted, unknowns are re-prompted (acceptable tradeoff vs. complexity of persisting)

**Why not SharedPreferences:** Cache entries are only valid for the current day. Persisting them would require a purge mechanism — more code, no real benefit.

## 7. Room Database

### 7.1 Why Room

Firestore writes require network. Agents frequently work in low-signal areas. Every detected call is written to Room first, then synced to Firestore when online. The My Calls tab always reads from Room — instant, no Firestore listener cost.

**Firestore listener cost saving:** Without Room, My Calls tab would need a real-time Firestore listener (\$\$). With Room, My Calls reads locally for free. Firestore is written to once per call (not listened to for display).

## 7.2 Schema

Entity: `CallLogEntity`

```
java

@Entity(tableName = "call_logs")
public class CallLogEntity {
    @PrimaryKey
    String dedupKey;           // number_timestamp/10000
    String agentUid;
    String agentName;
    String callerNumber;      // normalized 10-digit
    String contactName;       // "Unknown", "Personal Call", or contact name
    String callType;          // "incoming" | "outgoing"
    String callCategory;      // "office" | "personal" | "unknown"
    long timestamp;           // epoch millis
    long duration;            // seconds, 0 if missed
    boolean isMissed;
    boolean taskCreated;
    String taskId;            // null if no task
    boolean autoSaved;
    boolean syncedToFirestore; // false until confirmed write
}
```

## 7.3 Sync Strategy

Class: `CallLogSyncWorker.java` (`WorkManager`, `PeriodicWorkRequest`, 15-min interval)

```
Every 15 minutes (or on connectivity restored):
├─ Query Room: WHERE syncedToFirestore = false
│   └─ For each unsynced entry:
│       Write to Firestore call_logs/{dedupKey}
│       On success: UPDATE Room SET syncedToFirestore = true
```

**Cost note:** WorkManager 15-min periodic is the minimum Android allows. Each sync batch writes only unsynced entries — typically 0-2 writes per cycle during normal use.

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## 8. Firestore — New Collection

### 8.1 `call_logs` Collection

**Document ID:** `dedupKey` (same as Room primary key — idempotent writes)

```
call_logs/  
  {dedupKey}/  
    agentUid      : String  
    agentName     : String  
    callerNumber  : String  
    contactName   : String  
    callType      : String // "incoming" | "outgoing"  
    callCategory  : String // "office" | "personal" | "unknown"  
    timestamp     : Timestamp  
    duration      : long    // seconds  
    isMissed      : boolean  
    taskCreated   : boolean  
    taskId        : String  // null if none  
    autoSaved     : boolean
```

**Retention:** Permanent (Phase 6 reporting uses this collection)

**Firestore rules additions:**

```
match /call_logs/{logId} {  
  allow read: if isOwner() || resource.data.agentUid == request.auth.uid;  
  allow write: if request.auth.uid == resource.data.agentUid;  
  allow create: if request.auth != null;  
}
```

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## 9. Task Model Changes

### 9.1 New field in `Task.java`

```
java  
  
private boolean autoSaved; // true if saved silently during next call
```

### 9.2 Auto-save Flow

```
Agent has CallBottomSheet open (partially filled or not)  
New call detected  
  └─ Current sheet data → build Task object
```

```

        customerName : from sheet (or "Unknown" if empty)
        customerPhone : detected number
        serviceType   : default service type
        callType       : "office"
        status         : "pending"
        autoSaved      : true
    }
    └─ Write to Firestore silently
    └─ Dismiss current sheet
    └─ Increment autoSavedCount in TaskViewModel
    └─ Show badge on Tasks tab bottom nav
    └─ Open new sheet for new call

```

**Badge display:** Red dot on bottom nav Tasks icon, count shown as "2 tasks need review" banner at top of OfficeTasksFragment when autoSavedCount > 0.

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## 10. Bottom Sheet — CallBottomSheet

### 10.1 Layout: `bottom_sheet_call.xml`

|                                |                               |
|--------------------------------|-------------------------------|
|                                |                               |
| (drag handle)                  |                               |
| Incoming Call                  | ← or "Outgoing Call"          |
| Ravi Kumar • 9876543210        | ← contact name + number       |
| OFFICE TASK (giant button)     | ← one tap saves with defaults |
| Customer Name [Ravi Kumar ]    | ← pre-filled                  |
| Phone [9876543210 ]            | ← pre-filled, locked          |
| Service Type [Toner Filling ▼] | ← default, editable           |
| Description [ ]                | ← optional                    |
| [Save Task] [Skip]             |                               |
| ⌚ Auto-dismisses in 2:00       | ← countdown timer             |

**For unknown numbers:**





Unknown Number • 9876543210

[ OFFICE CALL ] [ Personal Call ] ← two big buttons first

(Office selected → expands form)

## 10.2 Behavior Rules

| Condition                          | Behavior                                       |
|------------------------------------|------------------------------------------------|
| Known office contact               | Sheet opens pre-filled, one-tap save available |
| Unknown number                     | Two-button category picker first, then form    |
| Personal cached number             | No sheet — log only                            |
| Agent taps "OFFICE TASK"           | Save with defaults, dismiss instantly          |
| Agent taps "Skip"                  | Log call, no task, dismiss                     |
| 2 minutes pass with no action      | Auto-dismiss, log as skipped                   |
| New call arrives while sheet open  | Auto-save current, open new sheet              |
| Agent picks "Personal" for unknown | Cache as personal, log only, no task           |
| Agent picks "Office" for unknown   | Add to office_contacts, show form              |

## 11. Missed Call Reminder

**Class:** `MissedCallReminderWorker.java`

**Type:** `WorkManager` `OneTimeWorkRequest` with 5-minute initial delay

**Triggered by:** `CallDetectionService` when `RINGING → IDLE` (missed) for office/unknown number

### Notification content:

Title : "Missed Office Call"

Body : "You missed a call from Ravi Kumar (9876543210)"  
or "You missed a call from Unknown (9876543210)"

Action: Tap → opens `CallBottomSheet` pre-filled for that number

**Dismissible:** Yes, swipe to dismiss, no action buttons.

**Constraint:** Only fires if `callCategory != "personal"`. Personal missed calls are silently logged, no notification.

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## 12. Contact Sync

### 12.1 First Launch Flow

```
Agent opens app after Phase 3 install
├─ PermissionActivity checks READ_CONTACTS + WRITE_CONTACTS
│   └─ User grants
│       └─ ContactSyncManager.syncOnFirstLaunch()
│           1. Read all contacts under "Phoenix Computers" label
│               from device ContactsContract
│           2. For each contact:
│               normalize phone number
│               check Room office_contacts mirror
│               if not found → write to Firestore office_contacts
│               if found (same number) → skip silently (conflict
resolution)
│           3. Show "X contacts synced to Phoenix Computers" toast
│           4. Set SharedPreferences flag: firstSyncDone = true
```

### 12.2 Creating the "Phoenix Computers" Group

**In-app guide shown on first launch:**

Set up Office Contacts

Before we can detect office calls,  
add your customers to a contact  
group on your phone.

1. Open Google Contacts
2. Tap Labels → Create label
3. Name it "Phoenix Computers"
4. Add your office customers

[Open Google Contacts] [Done →]

### 12.3 Auto-add New Office Numbers

When an agent marks an unknown number as "Office":

1. Create device contact under "Phoenix Computers" account  
(ContactsContract.RawContacts insert with ACCOUNT\_TYPE = "Phoenix Computers")
2. Write to Firestore office\_contacts
3. Update Room office\_contacts mirror
4. Update UnknownNumberCache: number → "office"

## 13. My Calls Tab — PersonalCallsFragment

### 13.1 Layout

|                                                                                                                                                |                        |
|------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| My Calls                                                                                                                                       | ← toolbar title        |
| [Date Filter ▼] [Agent ▼(owner)]                                                                                                               | ← filter bar           |
| Today, 14 Jun                                                                                                                                  | ← date group header    |
|  Ravi Kumar 2:34pm<br>9876543210 • 3m 12s<br>[Task Created ✓] |                        |
|  Missed Unknown 3:10pm<br>9123456789 • Missed<br>[No Task]  | ← red missed indicator |
|  Personal Call 4:05pm<br>Personal • 1m 44s                  | ← no name shown        |

### 13.2 Data Source

- **Reads from:** Room DB (`call_logs` table) — instant, no network cost
- **Owner view:** Reads from Firestore `call_logs` filtered by date/agent (one-time `.get()`, not a listener — cost saving)
- **Agent view:** Room only

## 13.3 Filters

| Filter                                 | Agent             | Owner |
|----------------------------------------|-------------------|-------|
| Date range                             | ✓                 | ✓     |
| Agent                                  | X (sees own only) | ✓     |
| Call category (office/personal/missed) | ✓                 | ✓     |
|                                        |                   |       |
|                                        |                   |       |

## 14. Permissions

### 14.1 Required Permissions (all mandatory)

xml

```
READ_PHONE_STATE      ← TelephonyManager, already in manifest
READ_CALL_LOG         ← CallLogPoller, already in manifest
READ_CONTACTS         ← Contact sync
WRITE_CONTACTS         ← Auto-add new office contacts to device
POST_NOTIFICATIONS    ← Missed call reminder, already in manifest
FOREGROUND_SERVICE    ← CallDetectionService, already in manifest
```

### 14.2 Permission Request Flow

```
MainActivity.onCreate()
├── PermissionManager.checkAll()
│   ├── Any missing?
│   │   └── Launch PermissionActivity (full-screen, non-dismissible)
│   │       Show rationale for each permission
│   │       [Grant Permissions] button → system dialogs
│   │       If all granted → proceed to app
│   │       If any denied → show "This permission is required" +
retry
│           If permanently denied → deep-link to app settings
```

**PermissionActivity is non-dismissible** — back button disabled, no skip option.

## 15. New Files Summary

### 15.1 New Java Files (12)

| File                                       | Package                 | Purpose                                        |
|--------------------------------------------|-------------------------|------------------------------------------------|
| <code>CallDetectionService.java</code>     | <code>utils</code>      | Foreground service, orchestrates all detection |
| <code>CallStateReceiver.java</code>        | <code>utils</code>      | PhoneStateListener wrapper (Layer 1)           |
| <code>CallLogPoller.java</code>            | <code>utils</code>      | CallLog.Calls reader (Layer 2 + outgoing)      |
| <code>ServiceWindowReceiver.java</code>    | <code>utils</code>      | AlarmManager start/stop receiver               |
| <code>CallScheduler.java</code>            | <code>utils</code>      | Schedules 8am/11pm alarms                      |
| <code>NumberResolver.java</code>           | <code>utils</code>      | Room → Firestore number lookup                 |
| <code>UnknownNumberCache.java</code>       | <code>utils</code>      | In-memory day cache for categorized numbers    |
| <code>MissedCallReminderWorker.java</code> | <code>utils</code>      | WorkManager 5-min missed call notification     |
| <code>CallLogSyncWorker.java</code>        | <code>utils</code>      | WorkManager Room → Firestore sync              |
| <code>ContactSyncManager.java</code>       | <code>utils</code>      | First-launch phone → Firestore contact sync    |
| <code>PersonalCallsFragment.java</code>    | root                    | My Calls tab UI                                |
| <code>CallBottomSheet.java</code>          | root                    | Bottom sheet during/after call                 |
| <code>CallLogViewModel.java</code>         | <code>viewmodels</code> | LiveData for My Calls tab                      |
| <code>PermissionActivity.java</code>       | root                    | Mandatory permission request screen            |

### 15.2 New Room Files (3)

| File                            | Purpose                                        |
|---------------------------------|------------------------------------------------|
| <code>CallLogEntity.java</code> | Room entity                                    |
| <code>CallLogDao.java</code>    | Room DAO (insert, query unsynced, mark synced) |
| <code>AppDatabase.java</code>   | Room database singleton                        |

## 15.3 New Model

| File                                                 | Purpose                                                    |
|------------------------------------------------------|------------------------------------------------------------|
| <code>CallLog.java</code> (in <code>models/</code> ) | Firestore POJO mirroring <code>call_logs</code> collection |

## 15.4 New Layouts (4)

| File                                     | Purpose                             |
|------------------------------------------|-------------------------------------|
| <code>activity_permission.xml</code>     | Full-screen permission rationale    |
| <code>fragment_personal_calls.xml</code> | My Calls tab                        |
| <code>bottom_sheet_call.xml</code>       | Call detection bottom sheet         |
| <code>item_call_log.xml</code>           | Single row in My Calls RecyclerView |

## 15.5 New Drawables (3)

| File                              | Purpose                |
|-----------------------------------|------------------------|
| <code>ic_call_incoming.xml</code> | Incoming call icon     |
| <code>ic_call_outgoing.xml</code> | Outgoing call icon     |
| <code>ic_call_missed.xml</code>   | Missed call icon (red) |

## 16. Modified Files

| File                                  | Change                                                                                                                    |
|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| <code>Task.java</code>                | Add <code>autoSaved</code> boolean field                                                                                  |
| <code>TaskViewModel.java</code>       | Add <code>autoSavedCount</code> LiveData                                                                                  |
| <code>FirestoreRepository.java</code> | Add <code>call_logs</code> CRUD methods                                                                                   |
| <code>MainActivity.java</code>        | Replace placeholder with <code>PersonalCallsFragment</code> , start <code>CallDetectionService</code> , check permissions |

| File                             | Change                                                                             |
|----------------------------------|------------------------------------------------------------------------------------|
| <code>BootReceiver.java</code>   | Add <code>CallScheduler.scheduleWindow()</code> call (Phase 1 stub already exists) |
| <code>AndroidManifest.xml</code> | Register new service, receiver, activity, Room DB                                  |

## 17. Firestore Cost Analysis

| Operation                             | Frequency                     | Reads/Writes per day (est.) |
|---------------------------------------|-------------------------------|-----------------------------|
| call_logs write (via sync worker)     | Per call, batched             | ~10 writes/agent/day        |
| office_contacts read (NumberResolver) | Only on Room miss             | ~2 reads/agent/day          |
| call_logs read (owner My Calls)       | On-demand <code>.get()</code> | ~5 reads/owner/day          |
| contact sync                          | First launch only             | 1 batch write               |

**Total estimated Firestore ops per agent per day: ~12 reads + 10 writes**

Well within Spark (free) tier limits for a team of up to 10 agents.

## 18. Build Order

Implementation will proceed in this exact order to ensure each file compiles before the next depends on it:

1. CallLog.java (model)
2. CallLogEntity.java + CallLogDao.java + AppDatabase.java (Room)
3. UnknownNumberCache.java
4. NumberResolver.java
5. FirestoreRepository.java (add call\_logs methods)
6. Task.java (add autoSaved field)
7. CallLogSyncWorker.java
8. MissedCallReminderWorker.java
9. CallStateReceiver.java
10. CallLogPoller.java
11. CallScheduler.java + ServiceWindowReceiver.java
12. CallDetectionService.java (orchestrates 1-11)
13. ContactSyncManager.java
14. PermissionActivity.java + activity\_permission.xml
15. CallBottomSheet.java + bottom\_sheet\_call.xml
16. CallLogViewModel.java
17. PersonalCallsFragment.java + fragment\_personal\_calls.xml + item\_call\_log.xml
18. MainActivity.java (modifications)
19. BootReceiver.java (modifications)
20. AndroidManifest.xml (modifications)
21. Firestore security rules update

## 19. Testing Checklist

| Test Case                                    | Expected Result                     |
|----------------------------------------------|-------------------------------------|
| Incoming office call, app in background      | Sheet appears during call           |
| Incoming unknown call → agent picks Office   | Contact added to Firestore + device |
| Incoming unknown call → agent picks Personal | Logged, no sheet next time today    |
| Same unknown calls again same day            | Defaults to previous choice         |
| Missed office call                           | Notification fires after 5 min      |
| Outgoing call to office number               | Sheet appears after call ends       |
| Call arrives while sheet is open             | Auto-save, new sheet opens          |
| Device offline during call                   | Logged to Room, synced later        |
| Boot at 9am                                  | Service starts automatically        |



| Test Case                                  | Expected Result                           |
|--------------------------------------------|-------------------------------------------|
| Boot at 11:30pm                            | Service does not start                    |
| Service killed by OEM at 10am              | Layer 2 poller catches next call          |
| Two agents call same number simultaneously | Each sees their own sheet independently   |
| Contact sync — duplicate number            | Firestore name wins, no duplicate written |

## 20. Open Items (Phase 9)

- Owner override for service window hours (8am-11pm → configurable)
- Service type intelligence (default to most frequent for repeat customer)
- Room DB purge policy (keep last 90 days, delete older)
- Voice note on bottom sheet (Phase 8 — Gemini transcription)

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*Document prepared for Phoenix Computers ServiceCallApp — Phase 3*  
*Ready for implementation on confirmation.*